

Project Kadence 2.0 - Alpha 3

Rolling Contingency Snapshot

Status: ACTIVE ALPHA 3 / M1 + M2 + PERSONA V2 ACCEPTED / M3 NEXT

Snapshot: 28 Aug 2026, Europe/London

Branch: kadence/2.0-alpha-3

Active Windows repo: C:\AI Project\Project-Kadence-2.0

Purpose

This is the Alpha 3 successor to the frozen Alpha 2 contingency. The Alpha 2 PDF remains historical and is not overwritten. This snapshot records current accepted Alpha 3 state, the local disk/repository state, the next robot-routing milestone, rollback rules and deliberately deferred work.

Current ground truth

The operator's pre-refresh inspection showed local HEAD and remote Alpha 3 HEAD both at 1b1518ff7ea0bdc016e63e19c65906b015bf9692. Documentation was then refreshed on the live branch.

- Milestone roadmap refresh: 3e642e7a16091dac89fef63219b95e9154a1ce4f
- Clean-slate handover: f5ae4f8f16bf3e583c35fc77006608dbdbd71965
- Rolling contingency Markdown source: 210215fbd65a00be4725d6eb7a0abf6ed15df36d

Preserved local working state

Do not reset, stash or delete these merely to make git status clean.

- M firmware/fetch_repos.py
- M firmware/tools/apply_m6_pixel_weather_display.py
- ?? experiments/xiaozhi-backend/.tools/
- ?? experiments/xiaozhi-backend/__pycache__
- ?? experiments/xiaozhi-backend/control_surface/KadenceControl-run-*.ps1
- ?? experiments/xiaozhi-backend/tmp/

Active Windows layout

- C:\AI Project\Project-Kadence-2.0 - active source tree
- C:\AI Project\Archive\... - historical only
- C:\AI Project\Kadence-Flash\M6-Pixel-Weather-995a255 - accepted M6 flash checkpoint
- ...\experiments\xiaozhi-backend\.runtime\local\ollama\models - LOCAL model store
- ...\experiments\xiaozhi-backend\.runtime\xiaozhi-esp32-server - pinned/materialized backend runtime
- ...\control_surface and ...\persona - active operator/persona sources

Frozen Alpha 2 baseline

- Frozen branch: kadence/2.0-alpha-2
- Closure head: c74d8949f33c6dea1d7df2bea248cad9e82d5dd1
- Final physical runtime: 348e7c0fc05a027ba9affc7677534e488bd338c9
- M6 backend validation: 6029c08cdcfbea6861daa4fb7b3cc7290a345569
- M6 pixel-weather firmware: 995a2556f42e030660d6ed651b782987ac4a3d8e
- Pinned Xiaozhi upstream: e1876f1ce19cad6e7bfd7c80e41dc56b2e858dd5
- Independent rollback: beta/project-kadence

Transport invariants - preserve during M3

- Xiaozhi v1 bidirectional WebSocket
- 16 kHz / 60 ms Opus robot uplink
- OpenAI Realtime ASR on accepted robot/LUNA path
- Windows Silero preferred endpointing with 700 ms sustained-silence hold
- ESP32 AFE fallback, 180 ms final Opus flush, 10 s hard capture cap
- Robot-owned microphone stop, final flush and playback lifecycle
- Sonia Edge TTS, accepted safe tool boundary and M6 read-only tools
- Accepted EYE geometry and weather-icon contract
- No model-selected arbitrary motion; retired M7 overlay stays retired
- No AUTO cognition routing; no silent provider fallback

A3-M1 - LOCAL standalone inference

CLOSED / PHYSICALLY ACCEPTED

- Project-owned Ollama runtime on Windows
- qwen3.5:4b, about 3.3 GB, context 8192
- 100 percent GPU placement on RTX 3060 Laptop 6 GB
- Factual and personality responses, clean stop/restart, no TCP 11434 leak
- Accepted checkpoint: a5af604eca1c356bcfe1094392c85f71e604543e

A3-M2 - Explicit LOCAL/LUNA Control Surface

CLOSED / PHYSICALLY ACCEPTED

- Explicit LOCAL/LUNA only; no AUTO
- START and STOP operate selected path only
- LOCAL and LUNA text chat with short multi-turn context
- Quiet Enter-to-send and explicit UTF-8 LUNA response decode
- LOCAL port-conflict failure proved no LUNA fallback
- LUNA port-conflict failure proved no LOCAL fallback
- Accepted closure: 06b3b61669fa3fc3a6041e73613dccffdbdbd63b
- Existing Control Surface EXE remains the normal launcher

A3-M2.1 - Canonical Kadence persona v2

CLOSED / PHYSICALLY ACCEPTED

Persona v2 was extensively exercised in ordinary conversation and accepted as the working personality baseline.

- bff15167dae8dfeea89157917e40d6375afe15b0 - tune persona v2
- b0cac02bcf912b0b50895213970bd6adf8039e11 - pin v2 for LOCAL
- 04b7a2a814346d01bbf294c3a30b28e1559b766c - restore process guard regex
- 1b1518ff7ea0bdc016e63e19c65906b015bf9692 - report identity v2

Canonical persona SHA-256: f70578920b8360db5a902f417cec426991ea88c0f632cd052b408b4301458166

- Companion-first rather than help-desk
- Calm contextual competence; precise and self-possessed
- Restrained dry wit and playful sarcasm
- Stable preferences/opinions for subjective questions
- Natural British English and compact spoken answers
- Selective warmth; no habitual AI/server/rack cliches
- Utility before performance
- No model-selected physical expression or motion

Freeze rule: do not keep retuning persona v2 while implementing M3. Reopen only for a specific repeatable defect or a deliberately scoped later persona/adaptation milestone.

NEXT: A3-M3 - Robot cognition integration

M3 is split into two independent physical gates. The key architectural rule is to create a server-side cognition-provider boundary rather than hard-wire Ollama into robot transport.

```
StackChan | | frozen Xiaozhi/audio transport v Kadence backend | +-- endpointing / ASR / session / TTS  
| +-- cognition provider +-- LOCAL -> Ollama / qwen3.5:4b +-- LUNA -> accepted remote Luna path
```

M3A - StackChan to LOCAL

Route a real spoken StackChan turn through LOCAL while preserving accepted transport, ASR, endpointing, session and TTS.

- Start deliberately in LOCAL robot mode
- StackChan connects normally with existing mic/listening lifecycle
- Real spoken transcript reaches LOCAL qwen3.5:4b
- Response is recognisably persona v2 and returns through existing TTS/robot playback
- Short follow-up context works
- Clean stop/restart; no port/process leak
- No hidden LUNA fallback and no transport regression
- No firmware change unless new physical evidence requires one

Read before editing

- ALPHA3_FRESH_CHAT_HANDBOVER_2026-08-28.md
- ALPHA3_MILESTONES.md
- M1/M2 validation documents
- Current persona v2 and local hash guard
- Current accepted Xiaozhi backend/provider code

First engineering question: Where is the narrowest server-side provider seam that lets the accepted LUNA LLM call be replaced by LOCAL Ollama for a turn without touching robot transport, ASR, endpointing or TTS?

Clean-slate rules

- One milestone at a time; one physical gate at a time
- Exact commit anchors
- No acceptance claim without operator evidence
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